Practice 06: Identifying Misleading Visual(s)

This PDF records the prompt-based conversation used to complete the HOME UMMoodle learning activity for Practice 06. It captures what was asked of the AI assistant and what the assistant produced.

Conversation metadata

Chat name: Practice 06 Identifying Misleading Visuals
Chosen model: Cursor Grok 4.6 (Cursor IDE)
Web environment: Cursor IDE on macOS (Safari/Chrome available)
Assignment date: 22 September 2026
Course: CISC7204 | Activity: Identifying Misleading Visual(s)
Prompt-instruction context: Use Alberto Cairo's three mechanisms from Graphics Lies, Misleading Visuals (hiding relevant data; displaying too much data and obscuring reality; distorting data through graphic forms). Prefer a published, APA-citable visual that is not already used as a worked example in Cairo's own chapter.

1. Why this conversation was started

The student needed help locating a real published visual, identifying the encoder's misleading mechanism(s), writing a concise forum post in English, citing the source in APA style, and packaging an annotated image plus this conversation record for the personal forum thread 'My Mislaeding Visual Identified'.

STUDENT | Opening prompt

I need to finish CISC7204 Practice 06 in English. Please help me locate a misleading visual that uses one or more of Cairo's mechanisms: (1) hiding relevant data, (2) displaying too much data and obscuring reality, and (3) distorting data through graphic forms. I also need an image I can upload, an APA citation and reference, and a concise explanation of how the encoder misleads the decoder.

AI ASSISTANT (Cursor Grok 4.6) | Response

Cairo, following Wainer, groups most graphic lies into three strategies: hiding relevant data to highlight what benefits the encoder; displaying too much data so the decoder cannot see the pattern; and using graphic forms in inappropriate ways so the picture distorts the numbers. A good example should have a published source, a clear visual trick, and enough public discussion that the numbers can be checked. I will avoid Cairo's own classroom examples (the Venezuelan election bars and the Spanish unemployment chart) and the overused inverted-axis Florida gun-deaths graphic, so the analysis is original to this activity.

STUDENT | Follow-up

Please propose a specific visual that has a stable URL and that can be cited in APA 7th edition. I want something where the mechanism is obvious once it is pointed out, but easy for a decoder to miss at first glance.

AI ASSISTANT (Cursor Grok 4.6) | Response

Use the Americans United for Life line chart that Rep. Jason Chaffetz projected during the 29 September 2015 House Oversight hearing with Planned Parenthood president Cecile Richards. The graphic is titled in the spirit of 'Abortions up -- life-saving procedures down.' It plots cancer screening and prevention services against abortions from 2006 to 2013. The printed numbers are real: about 2,007,371 cancer-related services and 289,750 abortions in 2006, versus 935,573 and 327,000 in 2013. The picture, however, is not. The encoder used an unlabeled dual y-axis so the smaller abortion series climbs above the larger cancer series. Alberto Cairo, the author of our assigned chapter, later told PolitiFact the graphic was 'a damn lie' and 'ethically wrong' (Qiu, 2015).

2. Identifying the mechanisms

STUDENT | Follow-up

Map this chart onto Cairo's three mechanisms. Which ones are actually used, and how does each one mislead the decoder? Please keep the explanation concise enough for a forum post.

AI ASSISTANT (Cursor Grok 4.6) | Response

Primary mechanism -- distorting data through graphic forms. The encoder maps two counts that share the same unit (number of services) onto two invisible and incompatible vertical scales. Position, the most powerful visual encoding, is therefore no longer proportional to quantity. A decoder who reads the crossing arrows, rather than the tiny endpoint labels, concludes that abortions overtook cancer screenings. They did not: 327,000 is still far below 935,573. Arrowheads add a second distortion by implying that the invented slopes will continue.

Secondary mechanism -- hiding relevant data. The encoder shows only two years and only two service types. Intermediate annual reports, STI/STD testing (which rose to about 4.5 million), and contraceptive services (about 3.6 million) are omitted. Vox notes that all non-abortion services together were almost unchanged: 10.29 million in 2006 and 10.26 million in 2013 (Lee, 2015). Cherry-picking the one declining 'life-saving' series hides that overall clinical volume did not collapse.

Not used -- displaying too much data. This chart is sparse, not crowded. Cairo's second strategy is the opposite problem.

STUDENT | Follow-up

Please give me APA 7th-edition references for the visual and for Cairo, plus a short in-text citation I can paste into the forum thread.

AI ASSISTANT (Cursor Grok 4.6) | Response

In-text citation for the articles that reproduce the visual:
(Chalabi, 2015; Lee, 2015; Qiu, 2015)

References

Cairo, A. (2015). Graphics lies, misleading visuals: Reflections on the challenges and pitfalls of evidence-driven visual communication. In D. Bihanic (Ed.), New challenges for data design (pp. 103-116). Springer. https://doi.org/10.1007/978-1-4471-6596-5_5

Lee, T. B. (2015, September 29). Whatever you think of Planned Parenthood, this is a terrible and dishonest chart. Vox. https://www.vox.com/2015/9/29/9417845/planned-parenthood-terrible-chart

Chalabi, M. (2015, October 2). The week in data: Probability, dating and deceiving charts. FiveThirtyEight. https://fivethirtyeight.com/features/the-week-in-data-probability-dating-and-deceiving-charts/

Qiu, L. (2015, October 1). Chart shown at Planned Parenthood hearing is misleading and 'ethically wrong.' PolitiFact. https://www.politifact.com/factchecks/2015/oct/01/jason-chaffetz/chart-shown-planned-parenthood-hearing-misleading/

3. Drafting the forum annotations

STUDENT | Follow-up

Write the forum post in English for a thread titled exactly 'My Mislaeding Visual Identified'. Include a brief description of the visual, annotations of my findings, the APA material, the mechanism explanation, and an AI declaration. Keep the tone academic but concise.

AI ASSISTANT (Cursor Grok 4.6) | Response

I prepared three files you can upload to the personal forum:
1) misleading_visual.png -- the published FiveThirtyEight comparison (Chalabi, 2015) of the AUL / Chaffetz chart beside an honest common-scale re-encoding of the same two endpoint values.
2) annotated_findings.png -- the same comparison plus a bar chart of the service categories the encoder hid.
3) this PDF -- the prompt conversation required by the activity.

The ready-to-paste forum text is saved as forum_post.md. The analysis stays on visual encoding. It does not argue a political position. That matches Cairo's own comment that the distortion is ethically wrong regardless of what one thinks about the underlying policy issue.

4. Result produced by the AI assistant

Located visual: Americans United for Life dual-axis line chart, displayed by Rep. Jason Chaffetz at a U.S. House Oversight hearing on 29 September 2015, and reproduced in Lee (2015) and Qiu (2015).

Mechanisms used: (1) distorting data through graphic forms -- unlabeled dual y-axis and arrowheads; (2) hiding relevant data -- only two years and two cherry-picked service types.

Mechanism not used: displaying too much data.

Key check for the decoder: 327,000 is smaller than 935,573, so the crossing lines are a property of the scales, not of the data.

5. Model declaration (end of conversation)

Did I use an AI LLM? Yes
Exact LLM used: Cursor Grok 4.6
Interface: Cursor IDE chat
Operating system / browser context: macOS; source articles checked at Vox and PolitiFact in a Chromium-based browser.
The student reviewed the sources, chose the visual, and is responsible for the wording uploaded to the HOME UMMoodle personal forum.